Fortuitous states of $\mathcal{N} = 4$ SYM at weak coupling:
results, code and cross-checks

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How this document and the results in it were produced. The computations, the code and this text were produced by Claude, Anthropic's AI model, working as AI agents for the six people named above. They set the questions, read and checked the outputs, and reviewed one another's results; a separate set of AI agents reviewed the results independently (section 9). Every number below either carries an automated certificate of the kind described in section 2 or is marked as unverified; none has been rederived by hand. The labels R-0xx (results) and T-0xx (computations) are those of the collaboration's records and are used in the comments of the code; appendix A lists each of them with the code, the data and the checks in this repository. This document supersedes the summary of 7 October 2026. Corrections are welcome.

Abstract

We summarise what the collaboration has established about the $\frac{1}{16}$-BPS cohomology of $\mathcal{N} = 4$ super Yang–Mills theory at finite rank: the classical Q-cohomology in the BMN sector for SU(2) through level $j = 44$, SU(3) through $j = 32$ and in one column at $j = 34$, SU(4) through $j = 30$, SO(5), SO(7), SO(9), SO(11), USp(6), USp(8), USp(10) and G_2; BMN indices for SU(5) and for SO($2N + 1$) against USp($2N$) up to $N = 6$; the full-letter cohomology of SO(7) and USp(6) at levels 18 to 22; and the perturbative fate of the lowest classes at one and two loops. The main results are: (i) the SU(3) threshold operator of Choi, Choi, Kim, Lee and Lee is Q-exact, and the eigenvalue criterion on which their non-graviton index rests is false; the first SU(3) non-graviton class is at $j = 30$ ($E = \frac{25}{2}$, $[3, 0]$), it is the brane-fused operator of de Mello Koch, Kim, Kim, Lee and Lee, it is odd under charge conjugation, and it survives one loop; every SU(3) black-hole class below $j = 34$ is fused on an SU(2) black hole by scalar branes, and at $j = 34$ the $[3, 1]$ column contains, besides a descendant of the fused class at $j = 32$, exactly one class that is not of that form; (ii) SU(2) has 20 non-graviton classes through $j = 44$, of which 5 lift at one loop; the threshold class has no admissible lifting partner at any order given a superconformal selection rule, d_2 vanishes on it by direct computation, and the index alone protects it at every coupling; (iii) SU(4) has no non-graviton class through $j = 30$; (iv) SO($2N + 1$) and USp($2N$) first differ at $j = 4N + 6$ in the representation $\mathrm{Sym}^{N-3}[2, 0]$ ($N = 3, \ldots, 6$); for $N = 3, 4, 5$ the difference is one fermionic fortuitous SO($2N + 1$) multiplet per irrep, the lowest-level fortuitous class of that group, the SO(7) class dressed by $N - 3$ separated branes, and lifted at one loop; (v) structural results: a charge-conjugation parity that makes every multi-graviton even, twisted and kernel-parity indices that detect black holes without a cohomology computation, multi-graviton classes invisible on the Coulomb branch, and a two-loop supercharge on derivative-free words fixed by nilpotency. We compare every overlapping statement with the published literature, list the cross-checks, and say what remains open. All code and data behind the results are in this repository.

Contents

1 Main results at a glance **3**

2 Setup, definitions and method **4**
2.1 Objects . . . 4
2.2 Gravitons and fortuity: the operational definition . . . 4
2.3 The highest-weight complex . . . 4
2.4 Certification . . . 5
2.5 Other tools . . . 5

3 SU(2) **6**

4 SU(3) **7**
4.1 Level 24: the published threshold is not a black hole . . . 7
4.2 Level 30: the first black hole . . . 8
4.3 Level 32: three classes, all fused . . . 9
4.4 Level 34: the first class beyond scalar brane fusion . . . 9
4.5 Beyond: what the indices force . . . 10
4.6 Perturbative fate of the first black hole . . . 10

5 SU(4) **and** SU(5) **11**

6 SO($2N+1$), USp($2N$), G_2 **11**
6.1 Index differences . . . 11
6.2 Direct cohomology at the first difference . . . 12
6.3 These are the lowest-level classes (R-018) . . . 12
6.4 SO(7), USp(6) and G_2 to level 28; SO(5) . . . 13
6.5 Brane fusion of the SO classes (R-016) . . . 13
6.6 One loop (R-020) . . . 13
6.7 Full letters: SO(7) and USp(6) at levels 18–22 (R-025) . . . 14
6.8 Kernel parity for SO, USp, G_2 (T-034) . . . 14

7 Structural results **14**
7.1 Charge-conjugation parity and index-only detection of black holes . . . 14
7.2 Non-Coulomb gravitons . . . 15
7.3 Brane fusion and cores . . . 15
7.4 Selection rules, and what the index protects . . . 16
7.5 The two-loop supercharge on derivative-free words . . . 16
7.6 Entropy-level lifting . . . 17

8 Comparison with the literature **17**

9 Cross-checks **19**

10 Open questions **20**

A Index of results, code and data **21**

B Reproduction **25**

1 Main results at a glance

Table 1 collects the lowest non-graviton classes found in the BMN sector. "Level" is the power j of t in the BMN index (section 2.1); a complete scan of all levels $j < j_0$ certifies the lowest *level*, while the lowest *energy* needs every level $j < 3E$, since $E \geq j/3$ in this sector.

group	lowest non-graviton class	E	complete scans	certified	quantum fate
SU(2)	$(5;4,4,4)$, $[0,0]$, $j=24$	$\frac{19}{2}$	$j \leq 44$	level, energy	survives 1 loop; all orders[a]; index[b]
SU(3)	$(5;7,4,4)$, $[3,0]$, $j=30$	$\frac{25}{2}$	$j \leq 32$	level	survives 1 loop; orders 3, 4[c]
SU(4)	none for $j \leq 30$	–	$j \leq 30$	level ≥ 32	–
SO(5)	none for $j \leq 28$	–	$j \leq 28$	level ≥ 30	–
SO(7)	$(1;3,3,3)$, $[0,0]$, $j=18$	$\frac{17}{2}$	$j \leq 28$	level, energy	lifted at 1 loop
SO(9)	$(1;5,3,3)$, $[2,0]$, $j=22$	$\frac{21}{2}$	$j \leq 22$	level	lifted at 1 loop
SO(11)	$(1;7,3,3)$, $[4,0]$; $(1;5,5,3)$, $[0,2]$; $j=26$	$\frac{25}{2}$	$j \leq 24$	level	lifted at 1 loop
USp(6)	none at $j = 20, 22, 24, 28$	–	$j = 20$–24, 28	–	–
G_2	none for $j \leq 28$	–	$j \leq 28$	level ≥ 30	–

Table 1: Lowest-level non-graviton classes in the BMN sector, sector labels $(2J; q_1, q_2, q_3)$ and $\mathrm{SU}(3)_R$ irreps $[q_1 - q_2, q_2 - q_3]$. [a]Given the superconformal selection rule of section 7.4. [b]Protected at every coupling by the index, given the SU(2) classification. [c]Order 4 uses the selection rule; order 2 is open (section 4.6). The USp(6) scans do not cover $j < 20$ or $j = 26$; USp(6) has no class in the sector $(1;3,3,3)$ at $j = 18$ either (R-005). For SO(9) and SO(11) the lowest energy is not established.

The statements that differ from the published literature are collected, with equation numbers, in section 8. The most consequential ones:

- The criterion "a multi-graviton polynomial that vanishes on diagonal matrices is Q-exact" (sec. 3 of [4]) is false. Counting gravitons on the diagonal undercounts them by the classes that vanish on the Coulomb branch, and the "non-graviton index" built on that count is not a detector of black holes (R-002, R-015).
- The SU(3) threshold operator, eq. (4.8) of [4], is Q-exact at $N = 3$; the $-t^{24}$ of their non-graviton index is a fermionic graviton class, represented by their own "trace relation" $R_{24}^{(0,3)}$ (R-009). The first SU(3) black hole is at $j = 30$ (R-007), the brane-fused operator O_{30}^3 of [6] (R-016); the first class not fused by scalar branes is at $j = 34$ (R-024).
- The SU(4) entries $-\chi_{[2,0]}t^{28}$ and $\chi_{[1,1]}t^{30}$ of eq. (3.15) of [4] are artifacts of the same count (R-008, R-014).
- The descendant factor printed in eqs. (3.7), (3.14), (3.15) of [4] should be $\prod_I (1 - t^2 x_I)$ (R-001); the thesis [5] has the correct factor.
- The operator O_{30}^3 of [6] needs three corrections to its printed form to be closed; with them its stated coefficient $-\frac{648}{5}$ is reproduced (R-016).

Everything else that we checked agrees with the literature, often to the last digit: the SU(2) operator of [3]; Table 2 of [4]; the SO(7)/USp(6) numbers of [7] and Table 1 of [8]; the SO(7) lifting and the rank 741 of [9]; the one-loop verdicts and the rank 912 of [11]; and the entropy excess 1.21% of [11].

2 Setup, definitions and method

2.1 Objects

We work with the classical (tree-level) supercharge $Q \equiv Q_0$ of Chang and Lin [1] acting on gauge-invariant words in the $\frac{1}{16}$-BPS letters, mostly in the BMN subsector with letters ϕ_m, ψ_m, f ($m = 1, 2, 3$), on which

$$Q\phi_m = 0, \qquad Q\psi_m = \epsilon_{mnp}\,\phi_n\phi_p, \qquad Qf = \textstyle\sum_m [\phi_m, \psi_m] \tag{1}$$

in the normalisation of the code (`code/fortuity/bmn/words.py`; the conventions are those of [1, 4], checked against the published counts R-001, R-005, R-007). A BMN sector is labelled $(2J; q_1, q_2, q_3)$ with $q_I = R_I + J$; the letters carry $\phi_I \to (0; e_I)$, $\psi_I \to (1; (1,1,1) - e_I)$, $f \to (2; (1,1,1))$. The *level* $j = 2\sum_I q_I$ is the power of t in the BMN index (fugacities $t^j x^{q_1-q_3} y^{q_3-q_2}$), the energy is $E = j/2 - J$, and the number of letters is $Y = \sum_I q_I - 2J$. Every letter has $2J \le j/3$, so $E \ge j/3$ at level j. Q commutes with $\mathrm{SU}(3)_R$ and lowers $2J$ by one. An $\mathrm{SU}(3)_R$ highest weight $\lambda = (q_1 \ge q_2 \ge q_3)$ labels a *column*, the irrep $[q_1 - q_2, q_2 - q_3]$ at all $2J$.

Outside the BMN sector we use the full set of letters (gaugini λ and derivatives) in the superfield conventions of [1, 10], with sectors $(2J_1, 2J_2; 2q_1, 2q_2, 2q_3)$ at letter number Y; the derivative grading d_{BMN} vanishes exactly on the BMN sector.

2.2 Gravitons and fortuity: the operational definition

A cohomology class is *multi-graviton* (monotone) if it lies in the span of products of single-trace cohomology generators, modulo Q-exact operators, decided by linear algebra with the letters taken as *generic* matrices of the gauge algebra (not diagonal ones). In the BMN sector of $\mathrm{SU}(N)$ the generators are the components u_n, v_n, w_n of $\mathrm{tr}\,\Phi^n$; we use n up to (at least) the top Casimir degree of the group (N for $\mathrm{SU}(N)$, $2N$ for $\mathrm{SO}(2N+1)$ and $\mathrm{USp}(2N)$, 6 for G_2), and larger sets where stated. The complement is called *non-graviton*; it is our operational form of "fortuitous" (black-hole candidate) in the sense of [1, 2]. Consistently, every non-graviton representative that was tested fails to be Q-closed at the next rank: the SU(3) class of R-007 at $N = 4, 5$, the SO(7) class on $\mathfrak{so}(9)$ (R-010), the SO(9) class on $\mathfrak{so}(11)$ and $\mathfrak{so}(13)$ (T-043), the SO(11) classes on $\mathfrak{so}(13)$ (R-020). (This tests a representative; a different representative closed at higher rank is excluded only by the graviton test itself.) We do *not* use the eigenvalue (diagonal-matrix) criterion of [4]; section 4.1 shows why. A class that is a multi-graviton but vanishes on Cartan-valued letters is called *non-Coulomb* (terminology of [9]).

2.3 The highest-weight complex

Since Q commutes with $\mathrm{SU}(3)_R$, the complex restricts to highest-weight vectors (the kernel of the raising operators E_1, E_2), and the cohomology of that subcomplex at $(2J; \lambda)$ is the multiplicity of the irrep λ in $H(Q)$ at that $2J$. The dimension m of each highest-weight space is known exactly from an n_f-refined Haar integral and the Weyl character formula. Highest-weight vectors are computed shape by shape in the free multi-trace space (E_i preserve the flavour-stripped shape of a word). Operators are evaluated on random matrices of the gauge algebra over $\mathbb{F}_p$, $p = 2^{26} - 5$, with fermions expanded in Grassmann frames (one frame per fermion number), and ranks are taken modulo p. The multiplicity of the irrep in cohomology is $h = m - r_{\mathrm{in}} - r_{\mathrm{out}}$; its multi-graviton part is the rank of the graviton highest-weight vectors modulo the exact subspace. Implementation: `code/fortuity/hwcoh` (driver `run_sectors.py`, parallel engine `sectorpar.py`, FLINT/BLAS kernels `ffblas.py`); groups $\mathrm{SU}(N)$, $\mathrm{SO}(2N+1)$, $\mathrm{USp}(2N)$ and G_2.

2.4 Certification

(a) *Per-sector self-check.* The rank of the highest-weight vectors at the chosen functionals must equal the exact multiplicity m, separately in every fermion degree. This verifies both that the functionals are faithful on the highest-weight space and that the word set (often restricted in trace length) spans it. With faithful functionals the ranks r and the graviton counts are exact over $\mathbb{F}_p$.

(b) *Euler characteristic.* In every column $\sum_{2J}(-1)^{2J}h$ equals the multiplicity in the exact index. The rank terms cancel in this sum, so it checks the dimensions and the bookkeeping, not the ranks.

(c) *Completeness by enumeration.* Every (column, $2J$) with a nonempty letter content is present.

(d) *Direction of errors.* A rank computed by random evaluation can only underestimate the true rank (Schwartz–Zippel [15, 16]). An underestimated rank can create a spurious non-graviton class, never hide one. Null results ("no non-graviton class") are therefore robust; positive results were recomputed with independent seeds and, where stated, with larger generator sets or independent engines.

(e) *Rational reconstruction.* Where exact rational coefficients are claimed, they were solved at several primes and combined by the Chinese remainder theorem [17, 18], then re-verified at fresh primes.

2.5 Other tools

- *Direct cohomology* on the full weight space or its Weyl-symmetric part: `code/fortuity/bmn/symdirect.py`, `direct.py` (R-001, R-002, R-005, R-006).
- *Exact indices* by Haar integration: `code/fortuity/bmn/fullindex.py` (SU(N)), `groupindex.py` (SO, USp, G_2), `fugacityindex.py` (character values at fixed fugacities).
- *Diagonal (eigenvalue) graviton counting* of [4, 14]: `code/fortuity/bmn/gravindex.py`.
- *Coulomb test*: rank of the graviton highest-weight vectors on Cartan-valued letters, where Q-exact operators vanish: `code/fortuity/hwcoh/noncoulomb.py`.
- *Block restriction* (partial Coulomb branch): `code/fortuity/bmn/blockrestrict.py`, `classify.py`, `code/fortuity/hwcoh/classreps.py`, `classreps_par.py`, `cores.py`.
- *Charge conjugation* $C : X \to -X^T$ and twisted indices: `code/fortuity/bmn/cparity.py`, `twistedindex.py`.
- *Full letters and one loop*: `code/fortuity/full` (letters λ and derivatives, SU, SO, USp; the one-loop supercharge Q_1 of Budzik and Kulp [10] in `q1.py`, with the `so`(n) completeness term; lifting tests in `lifting.py`; full-letter levels in `sector.py`, `levels.py`, `gravitons.py`). The parallel and GPU one-loop engine used for the large SU(2) and SU(3) tests is in `scripts/oneloop/su-threshold-classes` (`fullpar.py`, `gpu_rank.py`, `cpu_rank.py`).
- *Two loops*: the two-loop supercharge on derivative-free words, `q2bmn.py` in the same directory (section 7.5).

3 SU(2)

The threshold class (R-001). The first non-graviton class of SU(2) in the BMN sector is unique, at $j = 24$ in the sector $(5; 4, 4, 4)$ ($R_I = \frac{3}{2}$, $J = \frac{5}{2}$, $E = \frac{19}{2}$), as found in [1]; we call it O_0. The explicit operator of [3] was verified closed and not exact, and its $SU(1|3)$ descendants $Q_+^m O_0$ sit at $(6; 5, 4, 4)$ and permutations ($j = 26$, bosonic, $[1, 0]$) and $(7; 5, 5, 4)$ and permutations ($j = 28$, fermionic, $[0, 1]$). The supercharges Q_+^m carry weight $t^2 x_m$, so the descendant factor that multiplies the core of the non-graviton index is $\prod_I (1 - t^2 x_I)$, and with it the SU(2) core is exactly $-\sum_{n\geq 0} t^{24+12n}$. Three independent computations agree: direct cohomology with full 2×2 matrices in all 683 sectors with $j \leq 28$; the exact index (signed sums at every $j \leq 28$); and the eigenvalue graviton count, which equals the closed form (3.4) of [4] through $j = 44$.

Complete scan to $j = 44$ (T-027). All 446 $\mathrm{SU}(3)_R$ columns with $j \leq 44$ (3182 sectors, every nonempty $2J$) were computed with the highest-weight engine on a CPU cluster. Self-checks pass in every sector and the Euler characteristic equals the exact index in every column. The cohomology has 186 multiplets: 166 multi-graviton, all of Coulomb type (no non-Coulomb graviton in SU(2) through $j = 44$), and 20 non-graviton classes in 17 sectors:

j	24	26	28	30	32	34	36	38	40	42	44	total
non-graviton classes	1	1	1	1	1	2	3	2	2	3	3	20

All 20 have an odd number of letters, i.e. parity $P = -1$ (section 7.1), and in every column their number equals $-T$, the kernel parity index computed before the scan finished (R-017). The sectors are listed in `data/su2/bmn_hw_cohomology_j44.json` (assembled from the per-sector records by `fortuity.hwcoh.to_json`); the non-graviton counts of all 17 sectors were reproduced by the full-letter one-loop runs below (different code, different seeds).

One loop (T-040, T-037). Each of the 20 classes was tested with the one-loop supercharge Q_1 of [10]: its image is checked for Q_0-exactness in the partner sector $(2J - 1, 2J - 1; 2q)$ at $d_{\mathrm{BMN}} = 1$, with exact controls, two seeds, and a certificate that the words span. Five classes lift: the multiplet of O_1 (one class each at $j = 36$ $[0, 0]$, 38 $[1, 0]$, 40 $[0, 1]$, 42 $[0, 0]$) and, at $j = 44$ $[1, 0]$, one of the two classes of the sector $(11; 8, 7, 7)$, which has the charges of $w_2 O_1$. Fifteen survive: O_0 dressed by powers of w_2 and O_1', with their $\mathfrak{psu}(1, 2|3)$ descendants (O_0's multiplet at $j = 24$–30, $w_2 O_0$ at $j = 32$, O_1' at $j = 36$, $w_2^2 O_0$ at $j = 40$, $w_2 O_1'$ at $j = 44$, and the others listed in `scripts/oneloop/su-threshold-classes/runs/t040_oneloop_table.json`). This reproduces every verdict of [11] in its range (O_0, $w_2 O_0$, O_1' unlifted; O_1 lifted, with the exact-space rank 912 at $j = 36$) and extends it to $j = 44$. The verdicts had been predicted a priori from the selection rules of section 7.4 (T-037); the prediction was posted before the direct results and agrees in all 16 sectors it covers. Rank stability was checked at three times the number of evaluation points for the lifted sectors.

All orders for O_0. The order-n partner sectors of O_0, $(4, 4; 8, 8, 8)$ at $Y = 8 - n$ with $n \leq 4$, have cohomology $0, 27, 0, 4$ for $n = 1, 2, 3, 4$, all of Coulomb type. The Coulomb-type classes restrict injectively to the Cartan, where for SU(2) the module is the graded-symmetric algebra on one species of letters; that module has no lowest-weight vector of the weight of O_0 at $n = 2$ or 4. So, given the superconformal selection rule of section 7.4, O_0 has no admissible lifting partner at any perturbative order, with no Coulomb-branch assumption. Independently, the two-loop differential vanishes by direct computation: $d_2[O_0] = 0$ with the Q_2 of section 7.5, and likewise for the three descendants

of O_0 at $j = 26, 28, 30$ (`scripts/oneloop/su-threshold-classes/runs/su2_00_d2.json` and `t040_twoloop_table.json`). The two-loop runs for the other SU(2) survivors ($j \geq 32$) were not completed.

Index protection. No Coulomb-type multiplet with the weight of O_0 and the opposite statistics exists at any odd rung of its index tower (`scripts/oneloop/su-threshold-classes/su2_index_tower.py`, output `runs/su2_00_index_tower.txt`). Given the SU(2) classification of non-Coulomb classes at these charges (our BMN scans; a direct computation at rung +1; for the other rungs the results of [8], not rederived), the index alone protects O_0 at every coupling (section 7.4).

Higher survivors. For w_2O_0 and O_1' the order-2 and order-4 partner sectors contain only Coulomb-type classes ($h = 119$ and 470 for w_2O_0; 103 and 1123 for O_1', with 12 more at order 6) and order 5 is empty. At order 3 the partner sector of w_2O_0 has $h = 42{,}894 - 23{,}094 - 19{,}797 = 3$ classes, all non-Coulomb, so its stability at order 3 is open; the order-3 test of O_1' was not completed.

4 SU(3)

4.1 Level 24: the published threshold is not a black hole

All 29 classes are multi-gravitons (R-002). In the sector $(3; 4, 4, 4)$ ($R = \frac{5}{2}$, $J = \frac{3}{2}$, $E = \frac{21}{2}$) the Weyl-symmetric subspace has dimension 1898, 953 closed and 924 exact operators, hence 29 classes, as in Table 2 of [4]. The 807 symmetrised graviton monomials have rank 28 on diagonal matrices (the paper's "#gravitons = 28") but rank 29 modulo Q-exact operators with full 3×3 matrices, and $924 + 29 = 953$ saturates the closed space. Every class of the sector is therefore a multi-graviton. Explicit counterexample to the eigenvalue criterion: a 32-term polynomial $\sum c\, u_2 u_2 v_2 w_3$ with $c \in \{\pm 1, \pm 2, \pm 3\}$, Q-closed, identically zero on diagonal matrices and not exact (`data/su3/graviton_class_vanishing_on_diagonal_j24.json`; check: `python3 -m fortuity.bmn.r002_check`).

The threshold operator is exact (R-009). The operator $O = -6\, O_6^{(0,3)}$ of eq. (4.8) of [4] was built in two ways (from the printed expansion, and from the definitions through r_{10}, r_{14} with the traceless projections of their (A.3)); the two agree term by term (3747 canonical multi-trace terms). It is closed on traceless 3×3 matrices (not formally, not at $N = 4, 5$) and it is Q-exact at $N = 3$, by three independent computations (the Weyl-symmetric subspace at two primes, the full complex with certified functionals, and the highest-weight complex). An explicit preimage: an f-free SU(3)-singlet λ_0 in $(4; 4, 4, 4)$, a combination of 99 index-contracted multi-trace monomials with rational coefficients (common denominator 360), with $O = Q_{\text{code}}(\lambda_0) = i\, Q_{\text{paper}}(\lambda_0)$ on 3×3 traceless matrices (`data/su3/cckll_eq4p8_preimage_j24.json`; test `code/tests/test_r009_preimage.py`). The one-dimensional singlet cohomology of $(j, 2J) = (24, 3)$ is represented by the paper's own "fundamental trace relation" $R_{24}^{(0,3)} = \epsilon_{a_1a_2a_3} u^{a_1 b} v^{a_2}{}_b v^{a_3 c}{}_d v^d{}_c$ of its (A.4), which vanishes on diagonal matrices at $N = 3$ (not at $N = 4, 5$) but is not exact; $24R_{24}^{(0,3)} + \text{Sym}(P)$ is exact, with P the polynomial of R-002. Two errors cancel in the paper's Table 2: one graviton too few (28 instead of 29, from the diagonal count) and one exact operator called non-exact.

The list (A.4) of [4] (R-009). Of the entries for which the paper gives no Q-exact form, $R_{18}^{(0,2)}$, $(R_{20}^{(0,3)})^i$, $(R_{22}^{(0,3)})^{ij}$ and $(R_{30}^{(0,5)})^{ijk}$ are exact (genuine relations), while $R_{24}^{(0,3)}$, $(R_{26}^{(0,4)})^i$ and the singlet

$R_{30}^{(0,5)}$ are not: they represent the unique (graviton) classes of their sectors, 5 of the 287 components of (A.4). The entry of level 26 is the descendant of the one of level 24:

$$(R_{26}^{(0,4)})^1 = -\tfrac{1}{3}\, Q_+^1 R_{24}^{(0,3)} + Q(\cdots) \qquad (N=3) \tag{2}$$

in the paper's normalisation (`scripts/bmn/su3-level24/descendant_check.py`). In the sector of the $[3,0]$ entry (R-007's sector, four graviton classes and one non-graviton class) the paper's diagonal-vanishing polynomial is trivial, so the $-t^{30}[3,0]$ of the non-graviton index is the genuine class of R-007.

How the three entries vanish on the diagonal. On diagonal traceless 3×3 matrices the three eigenvalue vectors of the scalars span only a plane in flavour space, and flavour singlets see the fermions through four anticommuting pairings. As a result the charge sectors of $R_{24}^{(0,3)}$, $(R_{26}^{(0,4)})^i$ and the singlet $R_{30}^{(0,5)}$ contain *no* Weyl-invariant function of the eigenvalues at all (a Howe-duality count), so every operator of these sectors vanishes on the diagonal, graviton or not, exact or not; for the $[3,0]$ entry six such functions exist and the vanishing is a genuine condition (`scripts/bmn/su3-level24/diag_vanishing.py`; the counts are exact linear algebra on polynomials).

Table 1 of [4] revisited. With the descendant factor corrected to $\prod_I(1-t^2x_I)$, the paper's Table 1 (the core of the SU(3) non-graviton index) is reproduced entry by entry through $j=40$ from our diagonal counts (`table1_core.py`); with the printed factor the rows 26 and 28 would not be empty. Adding the parity P of section 7.1: the $[0,0]$ entries at $j=24$ and 30 are the graviton classes $R_{24}^{(0,3)}$ and $R_{30}^{(0,5)}$; every other entry through $j=40$ is P-odd, hence not a graviton (certified at $j\le 30$, and at $j=32$ by the scan below; conditional on the diagonal counts for $j=34$–40); the $[0,0]$ at $j=36$ and the $[1,0]$ at $j=38$ are P-even and undetermined. The bookkeeping of the first entry: in the singlet column at $j=24$ the exact index is $0=(-1)+(+1)$ from the fermionic $R_{24}^{(0,3)}$ at $2J=3$ and a bosonic graviton at $2J=4$; the diagonal count sees only the latter, so the paper's $Z-Z_{\rm grav}$ has -1, which it read as a fermionic black hole in $[0,0]$. It is the graviton $R_{24}^{(0,3)}$.

4.2 Level 30: the first black hole

Complete scan (R-007). All $\mathrm{SU}(3)_R$ columns with $j\le 30$ (172 columns, every $2J$) were computed: the first non-graviton class is unique, at $j=30$ in $[3,0]$ (highest weight $(7,4,4)$), $2J=5$, i.e. $R=(\frac{9}{2},\frac{3}{2},\frac{3}{2})$, $J=\frac{5}{2}$, $E=\frac{25}{2}$, fermionic, in a sector with four graviton classes. There is no non-graviton class at any $j\le 28$. Checks: self-checks in every sector, Euler characteristic = index in 172 of 172 columns, a second seed on the class and its neighbours, graviton generators up to $n=4$ and $n=5$ (T-021) with unchanged counts, and agreement with the direct engine on overlapping sectors.

Representative (T-021). An explicit highest-weight representative over $\mathbb{F}_p$ with 201,564 words, closed at $N=3$; Q of its lift is nonzero at $N=4,5$ (`scripts/bmn/su3-level30-representative`).

Brane fusion (R-016). Under the block restriction $\mathrm{SU}(3)\supset \mathrm{SU}(2)\times U(1)$ (a chain map for Q_0, section 7.3) the class restricts to $x_1^3\otimes O_0$, all other powers of x_1 being exact or zero. It is therefore the brane-fused operator O_{30}^3 of [6], eqs. (2.24)–(2.25), up to normalisation, gravitons and exact terms. We rebuilt their operator (`code/fortuity/bmn/dmk.py`; 6139 terms in its $[3,0]$

highest-weight component); three readings of the printed formulas are needed, each forced by a check: the mixed-index r and R tensors of their (A.2), (A.3) must be traceless parts; a Koszul sign is needed on the five terms with a fermion-odd prefactor vvv; and $r_{16}^{(0,2)}$ in the $\frac{8}{5}$ term must read $r_{16}^{(0,3)}$. With them the operator is closed at $N = 3$, not at $N = 4$, and its x_1^3 coefficient is $-\frac{648}{5}O_{24}^2$, as stated in their (2.25).

Charge conjugation (R-017). The class is odd under $C : X \to -X^T$, so $P = -1$ (two independent checks: via the restriction, and by diagonalising C in the sector). In its column the ordinary index is 0 while the C-twisted index is -10.

Lowest energy. The scans certify the lowest level. The lowest energy $E = \frac{25}{2}$ also needs the sectors with $E < \frac{25}{2}$ at higher level, i.e. $j = 34$ with $2J \geq 10$ and $j = 36$ with $2J = 12$; these small sectors were not computed (all of $j \leq 32$ is, see below).

4.3 Level 32: three classes, all fused

The complete scan of $j = 32$ (T-042; all 30 columns, 207 sectors, every $2J$, the settings of R-007) finds 760 multiplets, 757 multi-graviton and 3 non-graviton: $(6; 7, 5, 4)$ in $[2, 1]$ and $(5; 8, 4, 4)$, $(6; 8, 4, 4)$ in $[4, 0]$. Self-checks pass in every sector and the Euler characteristic equals the index in all 30 columns (re-verified when this repository was assembled). This is exactly the count forced by the kernel parity index of R-017 ($T = -1$ at $(7, 5, 4)$, $T = -2$ at $(8, 4, 4)$). All three classes are fused (R-024 (iv)): $(5; 8, 4, 4)$ has a non-graviton image $x^4 \otimes O_0$ (the operator O_{32}^3 of [6]), and $(6; 8, 4, 4)$, $(6; 7, 5, 4)$ have non-graviton images in their x^3 channels. By their charges the last two are the $(1; e_1)$ and $(1; e_2)$ descendants of the class of R-007. Data: `data/su3/hw_cohomology_j32_records`.

4.4 Level 34: the first class beyond scalar brane fusion

Counting (R-024). The complete column $\lambda = (8, 5, 4)$ ($[3, 1]$, $j = 34$, every $2J$) has 37 multiplets: 34 multi-graviton and 3 non-graviton, two at $(6; 8, 5, 4)$ ($E = 14$, 11 letters) and one at $(7; 8, 5, 4)$ ($E = \frac{27}{2}$). Under $\mathrm{SU}(3) \supset \mathrm{SU}(2) \times U(1)$ the image of a non-graviton class lies in $x^n \otimes$ (SU(2) non-graviton classes), equivariantly; the only channels with the right charges are $x^4\otimes$ (the SU(2) class at $(6; 5, 4, 4)$) and $x^3\otimes$ (the one at $(7; 5, 5, 4)$), so the fused capacity is one class at $2J = 6$ and one at $2J = 7$. Two classes at $2J = 6$ against capacity one force at least one class whose restrictions are all graviton or exact: a class not obtained by attaching scalar branes to an SU(2) black hole (the restriction maps of R-016 keep only the abelian components of the scalars; the abelian components of ψ and f are set to zero). Independently of the graviton counts, the C-twisted index gives exactly $(37 - 31)/2 = 3$ P-odd classes in the column, more than the total capacity 2.

Which class it is. Explicit representatives of all non-graviton classes at $j = 30$–34 were built and restricted in every capacity channel (`data/su3/core_restrictions_j30_j34.json`): the $2J = 7$ class and all three $j = 32$ classes are fused; at $(6; 8, 5, 4)$ the restriction map on the two classes has rank one, so exactly one combination has no scalar-brane channel. Its images have zero residue in both channels, re-checked on the combined expression; that class is a generic lift on 834,657 of the sector's 834,972 trace words, closed at fresh $N = 3$ points. With R-007 fused and the $j = 32$ classes fused, *every* SU(3) *black-hole class below* $j = 34$ *is scalar-fused, and the first class that is not sits at* $j = 34$, exactly one in this column. By charges, the fused $2J = 6$ class sits where the $(1; e_2)$ descendant of O_{32}^3 would, and the other one is a new $SU(1|3)$ primary (a reading by charges, not constructed). This is a statement about scalar branes only. A brane could also carry fermionic

components, and a restriction that keeps the abelian components of ψ would test whether this class is fused in that wider sense; that extension of the maps of R-016 has not been done, so whether any SU(3) black hole is free of brane structure altogether remains open.

4.5 Beyond: what the indices force

The kernel parity index T (section 7.1) gives lower bounds on the number of SU(3) black-hole multiplets: at least $1, 3, 5, 11, 19, 29$ at $j = 30, 32, \ldots, 40$ (and 45, 66 at $j = 42, 44$). These are certified at $j \leq 32$ by the direct scans, and checked in the one column $(8, 5, 4)$ at $j = 34$ (exactly 3 P-odd classes, graviton count $34 = D$); beyond that they are conditional on the diagonal Coulomb counts not being overcounted (very likely right, not certified). They include $T = -2$ in the $[n, 0]$ column at $j = 24 + 2n$ for $n = 4, \ldots, 8$, the tower of [6]. Independently of any Coulomb count, the C-twisted index alone forces at least 4 and 10 P-odd (hence non-graviton) classes in the columns $(9, 6, 6)$ at $j = 42$ and $(10, 6, 6)$ at $j = 44$.

4.6 Perturbative fate of the first black hole

The class of R-007, called R-007 below, sits at $(2J_1, 2J_2; 2q) = (5, 5; 14, 8, 8)$ with $Y = 10$ letters, $d_{\mathrm{BMN}} = 0$; it is C-odd, an $\mathrm{SU}(2)_R$ singlet and a lowest-weight vector of $\mathfrak{psu}(1, 2|3)$ (`scripts/oneloop/su-threshold-classes/runs/r007_lowest_weight.json`). The loop corrections Q_n have the charges of Q, change the letter number by $1 - n$ and raise d_{BMN} by n. Hence R-007 is never the target of a lifting, its order-n target lies in $(4, 4; 14, 8, 8)$ at $Y = 11 - n$, only $n \leq 4$ can occur, and the target must be C-odd.

- *Order 1* (T-031). Q_1 of the class is Q_0-exact: in the partner sector ($[3, 0]$ highest-weight multiplicity 145,197) the exact subspace has rank 76,781 at 145,475 functionals, and adding Q_1 of the class (or of any of the four gravitons of its sector) does not raise it; controls exact; two independent sets of points; GPU and CPU arithmetic. R-007 is not lifted at one loop.
- *Order 3.* The C-odd target cohomology is empty, $131{,}048 - 67{,}869 - 63{,}179 = 0$ (exact multiplicity from twisted Weyl integration; computed ranks are lower bounds, so the zero is rigorous). No lifting at order 3, with no assumption.
- *Order 4.* The target has 177 classes, all C-odd and all of Coulomb type, and none is a lowest-weight vector of the weight of R-007; excluded by the selection rule.
- *Order 2.* The C-odd target cohomology has dimension $218{,}008 - 112{,}730 - 103{,}379 = 1{,}899$, of which 1,898 are of Coulomb type. The one non-Coulomb class is not a lowest-weight vector (its $[2, 1]$ and $[2, 0]$ lowering images are not exact, with span certificates), so the selection rule excludes it. Eight Coulomb-type classes pass all lowest-weight tests; d_1 vanishes on them. They are excluded under the Coulomb-branch assumption (a Coulomb-type class is never a target), and otherwise only by d_2[R-007]. The direct computation of d_2[R-007] with the Q_2 of section 7.5 was set up on the cluster in two independent chains; it has not produced an exactness verdict, so order 2 is open.
- *Restriction argument* (T-033). Theorem A: every possible order-3 partner of R-007 is an SU(3) core (restricts to zero under $\mathrm{SU}(3) \supset \mathrm{SU}(2) \times U(1)$); classical, with all fused channels computed directly. The block restriction is not a Q_1 chain map (W-boson exchange), so it can prove lifting but not survival.

So R-007 is stable to all orders under the Coulomb-branch assumption, and without it at orders 1, 3 and 4. The index cannot do better (section 7.4): the eight multiplets above have the weight and

C-parity of R-007 and the opposite statistics, so its protection is equivalent to the Coulomb-branch statement for those eight classes. The all-letter index at the weight of R-007 is 35, its C-twisted version -565; without the all-letter graviton subtraction these decide nothing.

The non-Coulomb graviton $R_{24}^{(0,3)}$ at loop level. It can never be a target (grading), and it is not a source at one loop ($Q_1 R_{24}^{(0,3)}$ is Q_0-exact, exact-space rank 1897, two sets of points; T-030) nor at two loops (direct null test at rank 3 with the two-loop supercharge). The same pipeline reproduces the one-loop lifting of one of the two SU(2) classes at $j = 36$ [11].

5 SU(4) and SU(5)

- R-003. The diagonal (eigenvalue) count reproduces eq. (3.15) of [4]: $Z - Z_{\text{grav}} = -\chi_{[2,0]} t^{28} + \chi_{[1,1]} t^{30} + O(t^{32})$. For SU(5) the same difference vanishes through $j = 28$. No direct SU(5) scan was done.
- R-008, R-014. Complete highest-weight scans of all columns with $j \leq 28$ (145 columns) and $j = 30$ (27 columns), every $2J$: all cohomology is multi-graviton, so the SU(4) BMN threshold is at $j \geq 32$. Both terms of eq. (3.15) are artifacts of the diagonal count: the column $(6, 4, 4)$ ($[2, 0]$, $j = 28$) has $1, 2, 6, 30, 46, 19, 1$ classes at $2J = 0, \ldots, 6$ and the column $(6, 5, 4)$ ($[1, 1]$, $j = 30$) has $0, 3, 13, 36, 66, 50, 13, 1$ at $2J = 0, \ldots, 7$, all multi-graviton. Self-checks in every sector; Euler characteristic = index in 145/145 and 27/27 columns.
- Non-Coulomb gravitons (R-015): $(j, \lambda, 2J) = (28, (6, 4, 4), 3)$, $(30, (7, 4, 4), 4)$, $(30, (6, 5, 4), 4)$, $(30, (7, 4, 4), 3)$, one class each. The onset $j = 28$ is the "apparent threshold" of the diagonal index.
- Parity (R-017). Every SU(4) class through $j = 30$ is P-even. The C-twisted index, validated against the direct cohomology in all 172 columns with $j \leq 30$, is non-negative in every column through $j = 44$, so the indices force no SU(4) black hole in that range (`scripts/index/c-parity/runs/su4_twisted_scan_j44.json`).
- A fused candidate for the first SU(4) class ($x^3 y^3 \otimes O_0$ in $[6, 0]$ at $j = 36$, $2J = 5$, by analogy with the SO tower) was proposed but not computed; it would bound the threshold level from above only.

6 SO($2N + 1$), USp($2N$), G_2

6.1 Index differences

SO($2N + 1$) and USp($2N$) are exchanged by S-duality, and their BMN indices agree at low level. The first differences (R-004, R-021; exact Haar integrals, `code/fortuity/bmn/groupindex.py`, `fugacityindex.py`):

$$
\begin{aligned}
I_{\mathrm{SO}(7)} - I_{\mathrm{USp}(6)} &= -t^{18} + 3t^{20} - 3t^{22} + 11t^{24} - 33t^{26} + 39t^{28} - 74t^{30} + \ldots \\
I_{\mathrm{SO}(9)} - I_{\mathrm{USp}(8)} &= -6t^{22} + 18t^{24} - 33t^{26} + 117t^{28} - 289t^{30} + 564t^{32} + \ldots \\
I_{\mathrm{SO}(11)} - I_{\mathrm{USp}(10)} &= -21t^{26} + 63t^{28} - 154t^{30} + \ldots \\
I_{\mathrm{SO}(13)} - I_{\mathrm{USp}(12)} &= -56t^{30} + 168t^{32} + \ldots
\end{aligned}
$$

with $\mathrm{SU}(3)_R$ content of the first term $[0,0]$, $[2,0]$, $[4,0]+[0,2]$ and $[6,0]+[2,2]+[0,0]$ respectively, i.e. $-\mathrm{Sym}^{N-3}[2,0]$ (dimension $\binom{N+2}{5}$) at $j=4N+6$, the charge of one ψ and $2N+1$ scalars. For $N=3$ this reproduces [7]; $N=4,5,6$ are new as far as we know. For $N=6$ the content is established by character values at three random fugacity points plus the unrefined count (strong evidence, not a proof).

6.2 Direct cohomology at the first difference

group	sector	dim	closed	exact	classes	multi-graviton	non-graviton
SO(7)	$(1;3,3,3)$	903	683	559	124	123	1
USp(6)	$(1;3,3,3)$	903	682	559	123	123	0
SO(9)	$(1;3,3,3)$	936	699	559	140	140	0
USp(8)	$(1;3,3,3)$	936	699	559	140	140	0
SO(9)	$(1;5,3,3)$	5395	4133	3704	429	428	1
USp(8)	$(1;5,3,3)$	5395	4132	3704	428	428	0

(R-005, R-006; weight spaces, two seeds each.) The SO(7)/USp(6) rows reproduce Table 1 (BMN column) of [8] and sec. 4.3 of [7]; the SO(9)/USp(8) rows at 8 letters are a control. The SO(9) class ($\psi\phi^9$, $j=22$, $E=\frac{21}{2}$, highest weight of a $[2,0]$) is new as far as we know; the complete SO(9) scan of level 22 (16 columns, 79 sectors: 320 multiplets, 319 multi-graviton) reproduces it as the only non-graviton class at that level. For SO(11) (R-011, highest-weight columns at $j=26$, two seeds):

column	$2J$	SO(11): classes / graviton / non-graviton	USp(10)
$(7,3,3)$, $[4,0]$	0, 1, 2	0/0/0, 23/22/1, 0/0/0	0/0/0, 22/22/0, 0/0/0
$(5,5,3)$, $[0,2]$	0, 1, 2	0/0/0, 12/11/1, 0/0/0	0/0/0, 11/11/0, 0/0/0

So for $N=3,4,5$ the first index difference is exactly one fermionic non-graviton SO($2N+1$) multiplet per irrep at $2J=1$, absent in USp($2N$), with all graviton classes shared. The representatives fail to close one rank up: the SO(7) class (eq. (4.14) of [7]) on $\mathfrak{so}(9)$; the SO(9) representative on $\mathfrak{so}(11)$ and $\mathfrak{so}(13)$ (T-043; `scripts/bmn/so-usp-g2-scans/extensions/closure_check.py`); the SO(11) ones on $\mathfrak{so}(13)$.

6.3 These are the lowest-level classes (R-018)

Complete scans of every column and every $2J$ below the first difference:

group	scanned	columns	sectors	classes	non-graviton
SO(7)	$j\le 18$	52	170	166	1 (R-005's class)
SO(9)	$j\le 22$	82	313	665	1 (R-006's class)
SO(11)	$j\le 24$	101	416	1331	0

With SO(3) = SU(2) ($j=24$) and SO(5) (none for $j\le 28$), the lowest fortuitous level of SO($2N+1$) in the BMN sector is $j=24,\ge 30,18,22,26$ for $N=1,\dots,5$. For SO(7) the class is also the lowest in energy (all levels $j\le 28 > 3\cdot\frac{17}{2}$ are scanned); for SO(9) and SO(11) the energy statement is not established. Checks: self-checks in all sectors; Euler characteristic = exact refined index in every column; the SO(9) column $(5,3,3)$ at $j=22$ as a positive control (sector by sector $m=131/442/664/665/404/114/3$, one non-graviton class at $2J=1$); the audit script `scripts/bmn/so-odd-thresholds/analyze.py` reproduces the table from the data files.

6.4 SO(7), USp(6) and G_2 to level 28; SO(5)

- R-013, R-019. Complete scans of SO(7) and G_2 for $j = 20, \ldots, 28$ and of USp(6) for $j = 20, 22, 24$ and 28 ($j = 26$ was not scanned for USp(6)). SO(7) non-graviton classes: at $j = 20, 22, 24$ the $SU(1|3)$ descendants of the $j = 18$ class ($(2; 4, 3, 3)$, $(3; 4, 4, 3)$, $(4; 4, 4, 4)$); at $j = 26$ a new multiplet $(3; 5, 4, 4)$, $[1, 0]$, $E = \frac{23}{2}$, which is the product $O_{18}w_2$ of the $j = 18$ class with the graviton w_2 (a "hairy" class: the sector has exactly one non-graviton class and $O_{18}w_2$ is a nonzero non-graviton class there); at $j = 28$ two classes at $2J = 4$ in $(5, 5, 4)$ and $(6, 4, 4)$, with the charges of the descendants of the $j = 26$ class. Level totals at $j = 28$: SO(7) 1306 multiplets (2 non-graviton), USp(6) 1302, G_2 212 (all multi-graviton).
- USp(6) has no non-graviton class at $j = 20, 22, 24, 28$, nor in the sector $(1; 3, 3, 3)$ at $j = 18$ (R-005); $j < 20$ and $j = 26$ were not scanned.
- G_2 (R-010, R-022): multi-trace words span all G_2 invariants in every BMN sector with $j \leq 18$ (exact Haar dimension = rank of trace words); G_2 has no non-graviton class for $j \leq 28$ (complete scans; 83 classes for $j \leq 18$). The SO(7) class restricted to $\mathfrak{g}_2$-valued letters is a nonzero, non-exact *multi-graviton* class of G_2, and so is every SO(7) non-graviton class at $j = 22$–26. Caveat: the trace-word basis is proven complete for G_2 only for $j \leq 18$.
- SO(5) = USp(4) (R-012): complete scan of $j \leq 28$ (676 sectors, 960 classes), all multi-graviton, so the threshold level is ≥ 30 and any fortuitous class has $E \geq 10$. Four columns were recomputed with the USp(4) letters, identically.
- One SO(7) multi-graviton class without a USp(6) counterpart, at $(2; 6, 3, 3)$, $j = 24$, accounts for the $+[3, 0]t^{24}$ of the index difference; it vanishes on the Coulomb branch (R-013 (v)).

6.5 Brane fusion of the SO classes (R-016)

Under SO(9) $\supset$ SO(7) $\times$ SO(2) the SO(9) class restricts to $x_1^2 \otimes O_{\mathrm{SO}(7)}$ plus gravitons; under SO(11) $\supset$ SO(7) $\times$ SO(2)^2 (blocks x, y) the two SO(11) classes restrict to $x_1^2 y_1^2 \otimes O_{\mathrm{SO}(7)}$ ($[4, 0]$) and $(x_1 y_2 - x_2 y_1)^2 \otimes O_{\mathrm{SO}(7)}$ ($[0, 2]$; measured coefficient ratio $1 : -2 : 1$). The first SO($2N+1$)–USp($2N$) difference is the SO(7) black hole with $N - 3$ separated branes, each carrying one scalar bilinear ($\Delta j = 4$, $\Delta E = 2$), which accounts for $j = 4N + 6$, $E = 2N + \frac{5}{2}$ and the content $\mathrm{Sym}^{N-3}[2, 0]$; it also explains the absence of the tower in USp($2N$), which has no class at $j = 18$. This gives certificates of fortuity for the SO(9) and SO(11) classes that use only the SO(7) data.

6.6 One loop (R-020)

With the $\mathfrak{so}(n)$ version of Q_1 (`code/fortuity/full/q1.py`; checked against $Q_1 \operatorname{tr}(\phi_1 \psi_1) = 4(n-2) \operatorname{tr}(\lambda_+ \lambda_-)$ of [9] and the $\mathfrak{so}(n)$ completeness relation): the SO(7) class is lifted, $Q_1 O_{18}$ not lying in the span of the exact space (rank 741, the number of [9]) and Q_1 of the 324 graviton polynomials of its sector; the SO(9) class and both SO(11) classes are lifted as well, through their SO(7) core (one-way block tests with 1364, 4436 and 6458 graviton polynomials). In every sector tested, Q_1 of every graviton polynomial is Q_0-exact. So every SO($2N+1$) class at the first SO/USp difference, $N = 3, 4, 5$, is lifted at one loop, as S-duality requires (`python -m fortuity.full.lifting so7|so9|so11`; outputs in `scripts/oneloop/so-odd-classes/runs`). The block restriction commutes with Q_1 in cohomology on all three lifted fused classes (T-033; ratios -2, 288, -60 on both sides; `scripts/oneloop/protection`).

6.7 Full letters: SO(7) and USp(6) at levels 18–22 (R-025)

Complete highest-weight cohomology with all letters (gaugini and derivatives) at levels 18, 20, 22 (every column, every $(2J_1, 2J_2)$, every letter number):

- Level 18: SO(7) 985 multiplets, USp(6) 983. They differ only in the two sectors of Table 1 of [8], both reproduced in full: $(1,1;6,6,6)$, $Y=8$ (the BMN class: 903/559 exact/124 classes, $123+1$, against 903/559/123) and $(0,0;6,6,6)$, $Y=8$ (the chiral ring: 826/741/85 against 825/741/84, all multi-graviton; the extra SO(7) graviton vanishes on the Coulomb branch, as stated in [8]).
- Level 20: SO(7) 2588, USp(6) 2578; ten extra SO(7) multiplets in nine sectors, all at $Y=8$: five are non-graviton, five are $\mathfrak{so}(7)$-only multi-gravitons, all Coulomb-invisible.
- Level 22: SO(7) 6747, USp(6) 6713; 34 extra SO(7) multiplets in 19 sectors, all at $Y=8$; of the 21 in the twelve sectors that were classified, five are non-graviton and sixteen Coulomb-invisible multi-gravitons; 13 in the seven largest sectors are not classified.
- In every classified sector the Coulomb-type graviton counts of SO(7) and USp(6) agree, as S-duality requires. The derivative-free sectors agree column by column with the BMN engine.

6.8 Kernel parity for SO, USp, G_2 (T-034)

Here C acts trivially on invariants, so $P=(-1)^Y$. The kernel parity index (twisted index minus Coulomb count) for SO(7) to $j=30$ and USp(6), G_2 to $j=28$ is never negative: no black hole is forced by indices in these groups (their known classes have $P=+1$, which the index cannot see). Positive values are non-Coulomb gravitons where checked directly ($(8,3,3)$ at $j=28$ in SO(7) and USp(6); two G_2 sectors); elsewhere they rest on the signed Coulomb count (not certified).

7 Structural results

7.1 Charge-conjugation parity and index-only detection of black holes

Parity lemma (R-017). Charge conjugation $C: X \to -X^T$ on every letter commutes with Q. With Y the number of letters, every multi-graviton has $P := C(-1)^Y = +1$: single-trace gravitons are components of $\mathrm{tr}\,\Phi^n$, which have n letters and satisfy $\mathrm{tr}((-\Phi^T)^n) = (-1)^n \mathrm{tr}\,\Phi^n$, and both C and $(-1)^Y$ are multiplicative. Hence a class with $P=-1$ is automatically non-graviton. For groups on whose invariants C is inner or trivial (SU(2), SO($2N+1$), USp($2N$), G_2) the lemma says that every graviton has an even number of letters; it holds for all 4,475 graviton classes of these groups in our data. On Cartan-valued letters C acts as $(-1)^Y$, so every class visible on the Coulomb branch has $P=+1$.

Twisted index. $I_C = \mathrm{Tr}\,(-1)^{2J} C$ at fixed q equals $(-1)^{|q|}(\dim H^{P=+1} - \dim H^{P=-1})$, summed over $2J$ with no boson–fermion cancellation. Computed by twisted Weyl integration over the coset $\mathrm{SU}(N)\cdot C$ (`code/fortuity/bmn/twistedindex.py`), it agrees with the cohomology data in all 172 SU(3) and all 172 SU(4) columns with $j \le 30$. In the column of R-007 the ordinary index is 0 and $I_C = -10$.

Kernel parity index. With D the Coulomb-branch (diagonal) graviton count of a column, $T = (-1)^{|\lambda|} I_C - D$ is $\#\{P=+1\} - \#\{P=-1\}$ among the classes invisible on the Coulomb branch, so $T<0$ forces black holes. The independent review (section 9) showed that D, a six-term

signed Weyl alternation of randomly sampled dimensions, is not a certified lower bound; with that correction: the SU(2) values through $j = 44$ are exact (in the 17 forced columns $D = 0$, and every SU(2) diagonal dimension was also computed exactly over $\mathbb{Q}$); the SU(3) values are confirmed by direct cohomology at $j \leq 32$ and in the column $(8, 5, 4)$; beyond that they are very likely right but conditional. Q_1 preserves Y and commutes with C, so it conserves P: at one loop a P-odd class can pair only with another P-odd (non-graviton) class. This is why the P-odd threshold classes of SU(2) and SU(3) survive one loop, while the P-even SO(7) class is lifted by a graviton [9].

7.2 Non-Coulomb gravitons

Multi-graviton classes that vanish on Cartan-valued letters (R-015, R-023; complete lists in `data/noncoulomb`). Onsets in the BMN sector: SU(2) none through $j = 44$; SU(3) at $j = 24$ (the multiplet of $R_{24}^{(0,3)}$, verified explicitly: $R_{24}^{(0,3)}$ and $Q_+^1 R_{24}$, $Q_+^1 Q_+^2 R_{24}$, $Q_+^1 Q_+^2 Q_+^3 R_{24}$ are non-exact graviton classes at $(26, [1, 0], 4)$, $(28, [0, 1], 5)$, $(30, [0, 0], 6)$, plus a second primary, the singlet $R_{30}^{(0,5)}$ at $(30, [0, 0], 5)$); SU(4) at $j = 28$; SO(5) at $j = 20$; SO(7) exactly at $j = 24$; USp(6) at $j = 28$ (none at $j = 20$–24); none in SO(9) through $j = 22$ or SO(11) through $j = 24$. None occurs at $2J = 0, 1$ in the range computed. Properties:

- *S-duality.* The Coulomb-type graviton counts of $\mathrm{SO}(2N + 1)$ and $\mathrm{USp}(2N)$ agree in every sector compared (99 common sectors for $j \leq 24$, all of $j = 28$, and the classified full-letter sectors at levels 18–22), as their isomorphic Cartan data require; graviton mismatches between the two groups sit only in non-Coulomb classes.
- *Rank dependence.* $R_{24}^{(0,3)}$ vanishes on diagonal 3×3 matrices but not on 4×4 or 5×5 ones; the observation that Coulomb-invisibility is rank-specific, and the terminology, are due to [9]; the first non-Coulomb graviton is the SO(7) chiral-ring class of [8].
- *Counting.* The finite-N graviton count by "quantizing the moduli space" (eigenvalues of commuting diagonal letters, [14] sec. 6.1, used for the $\frac{1}{16}$-BPS sector in [4] sec. 3) undercounts the monotone spectrum by exactly these classes; $Z - Z_{\mathrm{grav}}$ computed that way is the index of the kernel of the Coulomb restriction, which contains non-Coulomb gravitons as well as black holes.
- *Quantum.* BMN classes are never targets of a lifting (section 7.4); Q_1 of every graviton polynomial tested is Q_0-exact, in all groups (R-020, T-030, T-031).

7.3 Brane fusion and cores

Evaluate an operator on block-diagonal letters in which some abelian blocks carry only scalar components x (separated branes). Since Q is built from commutators and the abelian parts are central, the coefficient of each monomial x^a defines a chain map from the complex of G to that of the unbroken non-abelian factor; it sends exact to exact and multi-gravitons to multi-gravitons. A non-graviton image therefore proves that a class is non-graviton and identifies the smaller-group class inside it (R-016). Results: the first SU(3) class is $x_1^3 \otimes O_0$, the first SO(9) and SO(11) classes are the SO(7) class dressed by one scalar bilinear per separated SO(2) block, and the first SU(3) class with no fused image is at $j = 34$ (R-024). In every case examined the fused classes share the one-loop fate of their core (the SO classes are lifted with the SO(7) core; the SU(3) class survives like O_0). The restriction is not a chain map for Q_1 (W-boson exchange), so it can prove lifting but not survival; the intertwining in cohomology was checked on three lifted classes (T-033). Reading: an SO(2) block is a D3-brane with its orientifold image; the SU(3) class is the SU(2) black hole

bound to one brane, the "dual-giant hair" of [6]. Thresholds read off as "first fortuitous class at rank N" therefore mix cores and fused classes.

7.4 Selection rules, and what the index protects

Gradings. Q_n changes the letter number by $1-n$ and raises d_{BMN} by n. So a BMN class is never the target of a lifting, the order-n partner of a class with Y letters has $Y+1-n$ letters, and only finitely many orders can occur for a given class. C is an exact symmetry of SU(N) $\mathcal{N}=4$ SYM and commutes with $Q(g)$: partners have the same C, and the order-n partner of a P-odd class has $P=(-1)^n$, so at odd n it cannot be a graviton.

Superconformal selection rule. The loop differentials commute with the tree-level generators of the commutant $\mathfrak{psu}(1,2|3)$ of Q; a lifting partner must then be a lowest-weight vector of the same weight. This follows from perturbative superconformal invariance (a framework hypothesis, not the Coulomb-branch assumption). Q_1 commutes exactly with all these generators except K and T; for those, the defect was checked to be Q_0-exact on the one-loop partner of the lifted SU(2) class at $j=36$. At two loops the rule predicts $d_2[O_0]=0$, which the direct computation confirms. It is argued and tested, not proven, beyond one loop.

The index. The BPS states at coupling g form a unitary lowest-weight module of $\mathfrak{psu}(1,2|3)$, whose Cartan charges are exactly the charges kept by the index. Since characters of distinct lowest-weight modules are independent, for each lowest weight μ the number $n_\mu=\sum_k(-1)^k a_k(\mu)$, with a_k the number of BPS multiplets of lowest weight μ at rung k (sectors $(2J-k,2J-k;2q)$), is independent of the coupling (for SU($N\geq 3$) separately for $C=\pm1$). Hence: at least $|n_\mu|$ multiplets of weight μ are BPS at every coupling; multiplets leave the BPS spectrum in pairs of the same μ and C and opposite statistics; and if no multiplet of weight μ exists at the rungs of one parity, all those at the other parity are BPS at every coupling. The last statement protects the SU(2) threshold class non-perturbatively (section 3) and fails for the SU(3) class, for which eight Coulomb-type multiplets of its weight and C-parity exist at rung $+1$ (section 4.6).

7.5 The two-loop supercharge on derivative-free words

$Q_1^2\neq 0$ on words (on 26 of 60 derivative-free test words), while nilpotency of $Q(g)$ requires $Q_1^2+\{Q_0,Q_2\}=0$ (eq. (12) of [10]), so a two-loop term is needed; [10] compute Q_1 only. We derive it on derivative-free words. In the superfield Feynman rules of the holomorphic twist the l-loop term of Q is a sum over Laman graphs with $l+2$ vertices, one of them the operator [10]. At two loops the only graph is K_4 minus an edge; with the operator at a two-valent vertex the three interaction vertices form a triangle and give $\delta^3(0)=0$, which leaves one graph, with three letters in and two out ($\Delta Y=-1$):

$$Q_2(C^A(\theta_1)C^B(\theta_2)C^C(\theta_3)) = K_2\, f_{APX} f_{BPQ} f_{CQY}\, \delta^3(\theta_1-\theta_2)\,\delta^3(\theta_2-\theta_3)\, D_2(C^X(\theta_1),C^Y(\theta_3)), \quad (3)$$

summed over the choice of the middle letter. On underived letters the kinematic factor must be an SU$(2)_R$ invariant with four derivatives, and there is exactly one, $D_2(f,g)=\epsilon^{ac}\epsilon^{bd}\,\partial_a\partial_b f\,\partial_c\partial_d g$. So on derivative-free words $Q_2=K_2U$ with U known, and nilpotency fixes the constant:

- $Q_1^2+K_2\{Q_0,U\}=0$ holds on 120 random derivative-free multi-traces (one or two traces of 2–5 letters) for $N=2,3,4$ with the single value $K_2=-\frac{1}{6}$ (Q_1 normalised as in `fortuity.full.q1`); $Q_1^2\neq 0$ on 54 of them, and $\{Q_0,U\}\neq 0$, so the value is unique;

- negative control: changing the weight of one term of D_2, no constant works on any of the 54 words;
- U commutes with $\mathrm{SU}(3)_R$ and C and anticommutes with $\partial/\partial\theta_a$ and S_a, as Q_1 does, and maps trace relations to relations at $N = 2$ and $N = 3$;
- eq. (2.19) of [11], $Q_1(Q_1O_0) = -Q_0(Q_2O_0)$, holds at rank 2 with this Q_2 and fails without it or with another constant.

Assumed: the true Q_2, in the scheme in which Q_1 is the operator above, has the graph form; within that form the solution is unique, and conjugations of $Q(g)$ change d_2 only by d_1-exact classes. Q_2 on letters with derivatives is not determined (it is not needed for d_2 of a derivative-free class). As far as we know this two-loop supercharge is not in the literature. Code: `scripts/oneloop/su-threshold-classes/q2bmn.py`, `test_q2.py`, `test_q2_tracerel.py`, `d2_make.py`.

7.6 Entropy-level lifting

- T-038. The bonus-charge refinement of the index in [11] implies that a fraction of the classical entropy must lift; at equal charges we reproduce their excess, $\delta_0 = 0.012134434$ (1.21%). Refining by P does not sharpen the bound: at fixed charges, J_R and d_{BMN} the letter-number parity is fixed, so P reduces to $\pm C(-1)^F$ and the P-graded index is the C-twisted one, whose Cardy growth cannot exceed the untwisted one (half of it for $\mathrm{SU}(N \geq 3)$, the same for groups on which C is trivial; for SO(7) we recover their ratio 1.0121).
- T-041. The charge dependence: over 927 points of the ordered wedge of charge space, δ_0 is largest at equal charges and decreases along every sampled ray; the supersymmetric black holes fill the disk $\sum_I \varphi_I^2 < 1$ in charge fractions, not just the triangle $\varphi_I > 0$; along the edge $Q_3 = 0$, δ_0 falls from 0.0100 at the midpoint to 0.0011 at fractions $(0.95, 0.05, 0)$. To first order in $\epsilon = (\dim G/j)^{1/3}$ at equal charges, $\delta = 0.0121 + 0.0551\,\epsilon + o(\epsilon)$, and the first-order coefficient is positive at every computed point. An independent code reproduces δ_0 to 5×10^{-10} and the first-order coefficient to 5×10^{-8} at 173 common points. The collapsed holonomy saddle dominates in all 432 verdicts (six groups, 18 charge points, four criteria) inside the triangle; dominance was not tested where a charge is negative. δ_0 bounds the lifted fraction only under the stated saddle-point assumptions.

8 Comparison with the literature

Paper by paper, every statement of ours that overlaps with a published one. "Agrees" means reproduced by our own computation, to the stated precision.

Chang, Lin [1]; [2].
Conventions, supercharge and the notion of fortuity. Agrees: the SU(2) BMN threshold at $j = 24$, $(5; 4, 4, 4)$, $E = \frac{19}{2}$, unique, with nothing below it in the BMN sector (R-001). The non-graviton representatives we tested fail to close at the next rank, as for fortuitous classes (section 2).

Choi, Kim, Lee, Park [3].
Agrees: the explicit SU(2) threshold operator is closed and not exact; its $SU(1|3)$ descendants are non-exact non-gravitons (R-001).

Choi, Choi, Kim, Lee, Lee [4].
The overlap is large; item by item:

- Table 2 (the $j = 24$ singlet sector: 1898 / 953 / 924 / 29): agrees exactly. Its split "28 gravitons + 1 black hole" does not: the 29 classes are all multi-gravitons (R-002).
- Sec. 3 and its appendix A, criterion "graviton polynomial vanishing on diagonal matrices $\Rightarrow$ Q-exact": false; explicit counterexample (R-002); in the three affected sectors of (A.4) there is no function on the diagonal at all (section 4.1).
- Eq. (4.8), the threshold representative: Q-exact at $N = 3$, explicit preimage (R-009). Their transcription and definition of the operator agree with each other (3747 terms); why their test reported it non-exact is not known.
- Eq. (3.4), the SU(2) graviton closed form: agrees through $j = 44$.
- Eqs. (3.7), (3.14), (3.15): the descendant factor $\prod_{I<J}(1-t^4x_Ix_J)$ is a misprint for $\prod_I(1-t^2x_I)$ (the thesis [5], eq. 6.33, has the correct one); with it the SU(2) core is exactly $-\sum_n t^{24+12n}$ and Table 1 is reproduced.
- Eq. (3.15), SU(4): the diagonal-count terms $-\chi_{[2,0]}t^{28}$, $\chi_{[1,1]}t^{30}$ are reproduced as diagonal-count statements and are artifacts: no non-graviton class in SU(4) for $j \le 30$ (R-003, R-008, R-014).
- List (A.4): R_{18}, R_{20}, R_{22} and $(R_{30})^{ijk}$ are genuine relations; $R_{24}^{(0,3)}$, $(R_{26}^{(0,4)})^i$ and the singlet $R_{30}^{(0,5)}$ are not, they are graviton classes, and $(R_{26}^{(0,4)})^i = -\frac{1}{3}Q_+^i R_{24}^{(0,3)}$ modulo exact terms (R-009).
- Table 1: reproduced with the corrected factor through $j = 40$ (signed). The $[0,0]$ entries at $j = 24$, 30 are gravitons; the $-t^{30}[3,0]$ is the genuine class of R-007; the other entries through $j = 40$ are P-odd; $[0,0]$ at $j = 36$ and $[1,0]$ at $j = 38$ are undetermined (section 4.1).

Lee, thesis [5].
Restates the diagonal criterion (same remark as above); eq. 6.33 has the correct descendant factor.

de Mello Koch, Kim, Kim, Lee, Lee [6].
Eqs. (2.24)–(2.25): with three corrections to the printed formulas (section 4.2) the operator O_{30}^3 is closed at $N = 3$ and its x_1^3 coefficient is $-\frac{648}{5}O_{24}^2$, as stated. It is the first SU(3) black hole (R-007, R-016); their next member O_{32}^3 is the $(5;8,4,4)$ class found by our $j = 32$ scan. The paper regards $j = 24$ as the SU(3) threshold (following [4]); there is no class there. Their picture of fused classes is confirmed below $j = 34$, where every SU(3) black hole is fused, and the first unfused core appears at $j = 34$ (R-024).

Gadde, Lee, Raj, Tomar [7].
Agrees: the closed forms of the SO(7) and USp(6) BMN indices (through $j = 18$, as far as their printed numerators determine them) and of the difference through $j = 30$; the singlet at $j = 18$; the numbers of sec. 4.3 (903 / 559 / 124 against 123); the representative eq. (4.14) is closed on $\mathfrak{so}(7)$ and not on $\mathfrak{so}(9)$. New relative to it: the SO(9), SO(11) classes and the index differences for $N = 4, 5, 6$ (R-004, R-006, R-011, R-021); the restriction to G_2 (R-010).

Chang, Lin [8].
Agrees: Table 1 in full, both columns and both groups, with the monotone/fortuitous split (BMN column 903 / 559 / 124 against 123; chiral-ring column 826 / 741 / 85 against 825 / 741 / 84), and the statement that the extra SO(7) chiral-ring graviton is invisible on the Coulomb branch (R-005, R-025).

Choi, Lee [9].
Agrees: the one-loop lifting of the SO(7) class, with the exact-space rank 741 in the target sector,

and $Q_1\mathrm{tr}(\phi_1\psi_1) = 4(n-2)\mathrm{tr}(\lambda_+\lambda_-)$ for $\mathfrak{so}(n)$ (R-020). Extended: the SO(9) and SO(11) classes are lifted too, through their SO(7) core.

Kim, Kim, Lee, Park [11].
Agrees: every one-loop verdict in our common range (O_0, w_2O_0, O_1' unlifted; O_1 lifted, exact-space rank 912 at $j = 36$); their eq. (2.19) with an explicit Q_2; the entropy excess 1.21% and the SO(7) ratio 1.0121. Extended: the complete SU(2) one-loop table through $j = 44$; for O_0, whose higher orders the paper leaves open (p. 6), no admissible partner at any order given the selection rule, $d_2[O_0] = 0$ directly, and protection at every coupling by the index (section 3); the charge dependence of the entropy excess (section 7.6).

Budzik, Kulp [10].
We use their one-loop supercharge Q_1 (with an $\mathfrak{so}(n)$ completeness term for the orthogonal groups) and their nilpotency relation eq. (12). The two-loop supercharge on derivative-free words with $K_2 = -\frac{1}{6}$ (section 7.5) is, as far as we know, new; [13] state for $\mathcal{N} = 1$ theories that $Q_1^2 \neq 0$ forces $Q_2 \neq 0$ and that a single two-loop diagram contributes, consistent with our counting.

Kinney, Maldacena, Minwalla, Raju [14].
Sec. 6.1: the eigenvalue count of finite-N gravitons misses the non-Coulomb gravitons (section 7.2). For the gaugino-free part of the $\frac{1}{8}$-BPS chiral ring that our BMN data cover, their statements are not contradicted in our range.

9 Cross-checks

check	where applied, and outcome
Exact-dimension self-check	Every sector of every highest-weight scan (all groups, BMN and full letters): rank of the highest-weight vectors at the functionals = exact multiplicity, per fermion degree.
Euler characteristic = exact index	Every column of every complete scan (SU(2) 446/446; SU(3) 172/172 at $j \leq 30$, 30/30 at $j = 32$, the $(8, 5, 4)$ column; SU(4) 145/145 and 27/27; SO(5), SO(7), SO(9), SO(11), USp(6), G_2 all columns).
Completeness by enumeration	Expected (column, $2J$) lists compared with the data files for all complete scans.
Second random seed	Every positive (non-graviton or non-Coulomb) count, and the one-loop tests (two seeds each); the SU(3) core column at $2J = 4$–7; identical results throughout.
Larger generator sets	SU(3) $j = 30$ with $n \leq 4, 5$; SO(7) with $n \leq 10$; unchanged counts.
Independent engines	Weight-space direct cohomology (`bmn.symdirect`) against the highest-weight engine (`hwcoh`) on SU(2) $j = 16$, SU(3) $j = 14$–18, SO(9) $j = 22$; full-letter engine (`full`) against `hwcoh` on all derivative-free sectors at levels 18, 20; the SU(2) non-graviton counts by the one-loop engine; SO(5) with USp(4) letters.
Literature anchors	Table 2 of [4]; eq. (3.4) of [4]; Table 1 of [8]; sec. 4.3 and the index closed forms of [7]; rank 741 of [9]; rank 912 and the one-loop verdicts of [11]; the 1.21% of [11]; $-\frac{648}{5}$ of [6].

check	where applied, and outcome
Exact arithmetic	Rational preimage of eq. (4.8) from five primes (CRT); diagonal-function counts as exact linear algebra; the SU(2) diagonal dimensions over $\mathbb{Q}$; exact multiplicities by Haar integration (rounding errors below 2×10^{-7}, results integral).
Controls	Exact controls $Q_0 u$ in every rank test; known-exact operators reported exact (e.g. $O_1^{(0,3)}$ of [4]); planted non-exact rows ("z+noise") reported non-exact; negative controls for K_2.
Rank stability	Doubling, and for the one-loop lifts tripling, of the evaluation points; spanning certificates.
Symmetries	$(2J_1, 2J_2) \leftrightarrow (2J_2, 2J_1)$ in every full-letter sector; $\mathrm{SU}(3)_R$ covariance of the block restriction coefficients $(1 : -2 : 1)$; Weyl symmetry of the diagonal dimensions.
S-duality	Equal Coulomb-type graviton counts for $\mathrm{SO}(2N+1)$ and $\mathrm{USp}(2N)$ in all compared sectors.
Index predictions	Kernel parity forecasts confirmed: SU(2) in all 446 columns; SU(3) at $j = 32$ and in $(8, 5, 4)$.
A-priori predictions	The SU(2) one-loop fate table predicted from selection rules before the direct computation (T-037); 16 of 16 sectors agree.
Independent review	The results R-001–R-018 were reviewed by a separate set of AI agents, with a cross-review of the claim list by an AI model from another developer (2026-10-01/02): the prose was consistent with the stored data and no stored number was wrong; the defects found (wording, claimed rigour, reproducibility) were corrected, including the status of the kernel parity index (section 7.1) and the energy bound $E \geq j/3$. Reproducer scripts: `scripts/review/independent-checks`.
Library tests	`code/tests`, run when this repository was assembled: 82 passed, 4 skipped (three slow tests that run with `FORTUITY_SLOW=1`, and one that needs `fork`, disabled on macOS).

What random evaluation proves, in one sentence: computed ranks never exceed true ranks over $\mathbb{F}_p$, so certified null results are exact over $\mathbb{F}_p$, and a bad prime could only add pairs of classes.

10 Open questions

1. d_2 of the SU(3) class R-007 (the only open order without the Coulomb-branch assumption); the validity of the superconformal selection rule beyond one loop; the lowest-energy status of R-007 (sectors $j = 34$, $2J \geq 10$ and $j = 36$, $2J = 12$).
2. The SU(2) survivors at higher order: order 3 of $w_2 O_0$ (three non-Coulomb classes in the target) and of O_1'; d_2 beyond the multiplet of O_0.
3. SU(3) cores: the column $(8, 6, 4)$ at $j = 36$ (fused capacity 0); the one-loop fate of the $j = 34$ core; the P-even entries $[0, 0]$ at $j = 36$ and $[1, 0]$ at $j = 38$ of Table 1 of [4] (a direct computation of the singlet column at $j = 36$ was started and not finished).

4. The SU(4) threshold ($j \geq 32$); SU(5) by direct cohomology.
5. SO(5) at $j = 30$; direct SO(13) cohomology at $j = 30$ (three fermionic classes predicted); SO(7) and USp(6) at level 30 (sizing done, scan not); USp(6) at $j = 26$; the 13 unclassified full-letter multiplets at level 22.
6. The C-twisted index with all letters at the charges of R-007, which is coupling-independent and could protect a C-odd state non-perturbatively.
7. Are non-Coulomb gravitons ever lifted as targets in SU(N)? Which fortuitous classes above the thresholds survive?
8. SO($2N$) groups (Pfaffian invariants), and the USp($2N$) thresholds beyond $j = 28$.

Note added. While this summary was being prepared we learned of the independent work of Kim, Lee, Lee and Song [12], whose results overlap with ours in SU(3) and SU(4). We thank them for sharing their results and for coordinating the posting.

A Index of results, code and data

Paths are relative to the root of this repository. `code/fortuity` is the library (with tests in `code/tests`); `scripts/<area>/<project>` hold the drivers, cluster job files and run outputs behind each result; `data` holds the results in plain formats, each file with a `.meta.json` stating what it contains, the code and command that produced it and how it was checked. "Labels" are those of the collaboration's records, which the code comments use.

label	content	code, data, checks
R-001	SU(2) threshold $j = 24$, descendants, descendant factor	`code/fortuity/bmn/direct.py`, `fullindex.py`, `gravindex.py`, `run_su2_direct.py`; `scripts/bmn/index-and-direct-cohomology` (`run_index.py`, `extract_core.py`, `check_su2_index.py`, `test_threshold_ops.py`); `data/su2/bmn_direct_cohomology_j28.json`, `bmn_index_j40.json`; check `python3 -m fortuity.bmn.run_su2_direct 28 OUT.json --check data/su2/bmn_direct_cohomology_j28.json`; `code/tests/test_result_drivers.py`.
R-002	SU(3) $j = 24$: 29 multi-graviton classes; criterion false	`code/fortuity/bmn/symdirect.py`, `r002_check.py`; `scripts/bmn/index-and-direct-cohomology` (`run_sym.py`, `test_diag_criterion.py 3,4,4,4`, `test_grav_sym.py`, `test_grav_robust.py`, `explicit_graviton_class.py`, `verify_explicit_class.py`); `data/su3/bmn_table2_weyl_symmetric_j24.json`, `graviton_class_vanishing_on_diagonal_j24.json`.
R-003	SU(4), SU(5) diagonal-count index	`code/fortuity/bmn/fullindex.py`, `gravindex.py`; `scripts/bmn/index-and-direct-cohomology` (`run_index.py`, `run_su4_energy.py`); `data/su4/bmn_index_j30.json`, `j28_highest_weight_sectors.json`, `data/su5/bmn_index_j28.json`.

label	content	code, data, checks
R-004	SO($2N+1$) vs USp($2N$) indices, $N = 3, 4, 5$	`code/fortuity/bmn/groupindex.py`; `data/so_sp/so7_sp3_bmn_index.json`, `so9_sp4_bmn_index.json`, `so11_sp5_bmn_index.json`; `code/tests/test_groupindex.py`.
R-005	SO(7)/USp(6) at $(1;3,3,3)$	`code/fortuity/bmn/symdirect.py`, `evaluate.py`; `data/so_sp/so7_direct_cohomology_1333.json`, `usp6_direct_cohomology_1333.json`.
R-006	SO(9) class at $(1;5,3,3)$	as R-005; `data/so_sp/so9_direct_cohomology_533.json`, `usp8_direct_cohomology_533.json`, `so9_hw_cohomology_j22.jsonl`, `so9_hw_cohomology_j22_seed2_533.jsonl`; closure test `scripts/bmn/so-usp-g2-scans/extensions/closure_check.py` (output `closure_check.txt`).
R-007	SU(3) complete scan $j \leq 30$; class at $j = 30$	`code/fortuity/hwcoh`; cluster jobs `scripts/bmn/hw-scans-cluster`; `data/su3/bmn_hw_cohomology_j30.json`; representative (T-021) `scripts/bmn/su3-level30-representative/representative.py` and `runs/`.
R-008	SU(4) $j \leq 28$	`code/fortuity/hwcoh`; `scripts/bmn/hw-scans-cluster`; `data/su4/bmn_hw_cohomology_j28.json`.
R-009	Eq. (4.8) of [4] exact; (A.4); diagonal mechanism; Table 1	`scripts/bmn/su3-level24` (`build_0.py`, `build_06.py`, `singlet_preimage*.py`, `hw_relation_test.py`, `triple_check.py`, `decompose*.py`, `full_complex_check.py`, `f_restricted_exactness.py`, `R24_check.py`, `descendant_check.py`, `diag_vanishing.py`, `table1_core.py`, `singlet_column_j36.py`); `data/su3/cckll_eq4p8_operator_j24.json`, `cckll_eq4p8_preimage_j24.json`, `cckll_A4_relations_exactness.json`; `code/tests/test_r009_preimage.py`.
R-010	G_2 completeness to $j = 18$; restriction of the SO(7) class	`code/fortuity/bmn/g2_checks.py` (`python3 -m fortuity.bmn.g2_checks restriction`, `completeness 18`); `data/g2/g2_so7_restriction_1333.json`; `code/tests/test_g2_checks.py`.
R-011	SO(11)/USp(10) at $j = 26$	`code/fortuity/hwcoh` (groups `SO_odd`, `Sp`); job files `scripts/bmn/so-usp-g2-scans/cluster/t020_*.sbatch`; `data/so_sp/so11_hw_cohomology_j26*.jsonl`, `usp10_hw_cohomology_j26*.jsonl`.
R-012	SO(5) none for $j \leq 28$	`code/fortuity/hwcoh`; `scripts/bmn/so5-scan` (`run_so5.sh`, `check_so5_driver.py`); `scripts/bmn/hw-scans-cluster` (`hw_par.sbatch`, `so5_json.py`); `data/so5/bmn_hw_cohomology_j28.json` (and the $j \leq 24$, $j \leq 26$ snapshots).
R-013	SO(7), G_2, USp(6) at $j = 20$–26	`code/fortuity/hwcoh` (`restrict.py`, `evalg.py`); `scripts/bmn/so-usp-g2-scans` (`restrict_scan.py`, `euler_check.py`, `extra_checks.py`, `cluster/t023_*.sbatch`); `data/g2` (files `*_j20*`, `*_j22_j24*`, `*_j26*`, `so7_robustness_seed2_gen10.jsonl`, `extra_564448`).
R-014	SU(4) $j = 30$	`code/fortuity/hwcoh`; `scripts/bmn/hw-scans-cluster/cluster/launch_su4_j30.sbatch`; `data/su4/bmn_hw_cohomology_j30.json`.

label	content	code, data, checks
R-015	Non-Coulomb gravitons	code/fortuity/hwcoh/noncoulomb.py (python -m fortuity.hwcoh.noncoulomb GROUP N GEN_NMAX OUT DATA...); data/noncoulomb (README.meta.json); code/tests/test_noncoulomb.py.
R-016	Block restriction, fused classes, the operator of [6]	code/fortuity/bmn/blockrestrict.py, classify.py, dmk.py, code/fortuity/hwcoh/classreps.py; scripts/bmn/block-restriction/code (run_su3_to_su2.py, run_so_tower.py, make_r016_data.py) and runs/; data/su3/r007_block_restriction_su2xu1.json, r007_x1cubed_image_su2.json, dmk_o330_hw.json, data/so_sp/*block_restriction*.json, *nongraviton_rep*.json; code/tests/test_blockrestrict.py, test_dmk.py.
R-017	Parity lemma, twisted and kernel parity indices	code/fortuity/bmn/twistedindex.py, cparity.py, gravindex.py; scripts/index/c-parity/code (twisted_index.py, noncoulomb_parity.py, su2_predictions.py, diag_counts.py, check_r007_cparity.py, atlas.py, su4_twisted_scan.py) and runs/; the review of its rigour in scripts/index/c-parity/c19; data/su3/twisted_index_j30.json, data/su4/twisted_index_j30.json, data/su3/kernel_parity_j40.json, data/su2/kernel_parity_j44.json; code/tests/test_twistedindex.py.
R-018	SO(7), SO(9), SO(11) lowest levels certified	code/fortuity/hwcoh; scripts/bmn/so-odd-thresholds (make_tasks.py, dims.sbatch, col.sbatch, euler.sbatch, audit analyze.py); data/so_sp/so7_hw_cohomology_j18.jsonl, so9_hw_cohomology_j20.jsonl, so9_hw_cohomology_j22.jsonl, so11_hw_cohomology_j24.jsonl, so9_hw_cohomology_533_j22_control.jsonl.
R-019	SO(7), USp(6), G_2 at $j = 28$	code/fortuity/hwcoh; scripts/bmn/so-usp-g2-scans (make_sector_tasks.py, extra_checks.py, cluster/t023_scan_v2.sbatch, cluster/t023_j28_checks.sbatch); data/g2/*_j28.jsonl, data/g2/checks_j28.
R-020	One-loop lifting of the SO($2N + 1$) classes	code/fortuity/full/q1.py, lifting.py (python -m fortuity.full.lifting so7|so9|so11); scripts/oneloop/so-odd-classes (development scripts and runs/); code/tests/test_q1_groups.py, test_lifting.py.
R-021	SO(13) vs USp(12) index	code/fortuity/bmn/groupindex.py, fugacityindex.py; data/so_sp/so13_sp6_bmn_index.json; log scripts/bmn/so-usp-g2-scans/extensions/n6_j30_fugacity.txt; code/tests/test_fugacityindex.py.
R-022	G_2 $j \leq 18$	code/fortuity/hwcoh (group G2); data/g2/g2_hw_cohomology_j18.jsonl; column list and log scripts/bmn/so-usp-g2-scans/extensions/g2.
R-023	Non-Coulomb gravitons, SO series and $j = 28$	code/fortuity/hwcoh/noncoulomb.py; data/noncoulomb/noncoulomb_*_j18|j22|j24|j28.jsonl, so_odd_j28_extension.meta.json, logs_2026-10-06.

label	content	code, data, checks
R-024	SU(3) core at $j = 34$	`code/fortuity/hwcoh` (`run_sectors.py`, `cores.py`, `classreps_par.py`); `scripts/bmn/su3-cores` (`cluster/sector.sbatch`, `analysis/make_core_data.py`, `reps/run_reps.py`, `reps/reps.sbatch`, `reps/make_reps_data.py`) and its `results*`; `data/su3/hw_cohomology_8_5_4.json`, `core_restrictions_j30_j34.json`; `code/tests/test_cores.py`, `test_classreps_par.py`.
R-025	Full letters, SO(7)/USp(6) levels 18–22	`code/fortuity/full` (`sector.py`, `gdims.py`, `levels.py`, `gravitons.py`); `scripts/full-letter/so7-usp6` (`enum_level.py`, `sector_group.py`, `hwgrav.py`, `runs/*.py`); `data/fullso` (`fullso.meta.json`); `code/tests/test_full_groups.py`.
T-019	One loop for SO classes (= R-020)	see R-020.
T-021	Explicit representative of R-007	see R-007.
T-025	SO thresholds (= R-018)	see R-018.
T-027	SU(2) complete scan $j \leq 44$	`code/fortuity/hwcoh`; `scripts/bmn/su2-level44-scan` (`setup.sbatch`, `su2_scan.sbatch`, `su2_col.sbatch`, `seed_columns.py`, `watch_su2*.sbatch`, `tasks_su2_j44.txt`); per-sector records `data/su2/hw_cohomology_j44_records`, assembled `data/su2/bmn_hw_cohomology_j44.json` (`python -m fortuity.hwcoh.to_json`); non-Coulomb test `data/noncoulomb/noncoulomb_su2_j44.jsonl`.
T-030, T-031, T-040	One loop, all orders and two loops for the SU(2) and SU(3) classes	`scripts/oneloop/su-threshold-classes`: `oneloop_test.py`, `fullpar.py`, `gpu_rank.py`, `cpu_rank.py` (one loop); `c_parity.py`, `twisted_index.py` (C); `size_r007_orders.py`, `membership.py`, `hwcheck.py`, `abelian_lw*.py`, `su2_lw.py`, `test_q1_symm.py` (orders 2–4, selection rule); `su2_index_tower.py` (index); `q2bmn.py`, `test_q2.py`, `test_q2_tracerel.py`, `d2_make.py`, `d2_report.py` (two loops); `cluster/` (job files); `runs/` (outputs, including `t040_oneloop_table.json`, `t040_twoloop_table.json`, `r007_*`).
T-032	SU(3) cores (= R-024)	see R-024.
T-033	Restriction and all-orders protection	`scripts/oneloop/protection/code` (`so9_intertwining.py`, `so11_intertwining.py`, `su2_channels.py`) and `runs/`.
T-034	Kernel parity for SO, USp, G_2	`scripts/index/kernel-parity-so-usp-g2/kernel_index.py` and its outputs.
T-037	A-priori SU(2) fate table	`scripts/oneloop/su2-fate-predictions/scripts` (`run_all.sh`) and `outputs/`.
T-038	Bonus index with parity	`scripts/entropy/bonus-index-parity` (`cardy_lib.py`, `cardy_validate.py`, `cardy_kklp.py`, ..., outputs `*.out`).
T-041	Charge dependence of the lifting excess	`scripts/entropy/lifting-excess` (`code/`, `physics/`, independent implementation `indep/`, `cluster/`, `outputs/`, `figures/`).

label	content	code, data, checks
T-042	SU(3) complete scan $j = 32$	`code/fortuity/hwcoh`; `scripts/bmn/su3-level32-scan` (`dims.sbatch`, `col.sbatch`, task lists); `data/su3/hw_cohomology_j32_records`; Euler check `python -m fortuity.hwcoh.to_json 'data/su3/hw_cohomology_j32_records/*.jsonl' 3 OUT.json`.
T-043	SO(13) index, G_2 below $j = 20$, SO(9) closure	`scripts/bmn/so-usp-g2-scans/extensions`.
T-044–T-048	SO(9) level 22, corrections, non-Coulomb extension, library promotion	see R-018, R-019, R-023, R-025.
other	labels in code comments	T-020 (computations of R-004, R-011), T-023 (R-010, R-013, R-019, R-022), T-024 (R-012), T-028 (R-016), T-029 (R-017); T-016, T-039: promotion of the full-letter code and of the $\mathfrak{so}(n)$ one-loop supercharge into `code/fortuity/full`; T-001, T-010: conventions and the operational definition (section 2).
D-005	Operational graviton definition	section 2.
D-006	BMN conventions	section 2.1; `code/fortuity/bmn/words.py`.
other	Validation of the full-letter code	`scripts/full-letter/su-checks` (`validate_su2.py`, `validate_hw_su2.py`, `run_bmn_check.py`, `classes_full.py`, `test_q1.py`).
other	Independent review	`scripts/review/independent-checks` (`reproducers/`, `followup/`).

B Reproduction

Installation. Python ≥ 3.10 with numpy, scipy, sympy (`code/pyproject.toml`):

```
python3 -m pip install -e 'code[dev]'   # or: export PYTHONPATH=$PWD/code
cd code && python3 -m pytest -q         # FORTUITY_SLOW=1 also runs the slow tests
```

The highest-weight engine runs on numpy alone; the optional FLINT/BLAS kernels are built with the scripts in `code/fortuity/hwcoh/flint` (FLINT 3, OpenBLAS or MKL). The GPU one-loop ranks need CuPy.

Quick checks (from the repository root unless stated; times on an Apple-silicon laptop).

```
cd code
python3 -m fortuity.bmn.r002_check              # R-002 operator, seconds
python3 -m fortuity.bmn.g2_checks restriction   # R-010, ~20 s
python3 -m fortuity.full.lifting so7            # R-020, SO(7) lifting
python3 -m fortuity.bmn.run_su2_direct 28 /tmp/su2.json \
        --check ../data/su2/bmn_direct_cohomology_j28.json   # R-001, ~8 min
cd ..
PYTHONPATH=code python3 -m fortuity.hwcoh.to_json \
        'data/su3/hw_cohomology_j32_records/*.jsonl' 3 out.json   # T-042 Euler check
export PYTHONPATH=$PWD/code REPO=$PWD
python3 scripts/bmn/su3-level24/table1_core.py   # Table 1 of CCKLL, seconds
cd scripts/bmn/su3-level24
python3 diag_vanishing.py       # diagonal mechanism, seconds
```

```
python3 descendant_check.py 1   # R_26 = -1/3 Q_+ R_24 + exact, ~1 min
cd ../so-odd-thresholds
python3 analyze.py 5 24 ../../../data/so_sp/so11_hw_cohomology_j24.jsonl
cd ../../oneloop/su-threshold-classes
python3 test_q2.py 12           # K_2 = -1/6, seconds
python3 su2_index_tower.py      # index tower of O_0, ~10 min
cd ../../entropy/bonus-index-parity && python3 cardy_validate.py
cd ../../full-letter/su-checks && python3 validate_su2.py 6
```

Scripts that are not modules need `PYTHONPATH` to contain `code`; scripts under `scripts/oneloop/su2-fate-predictions` also read the repository root from `REPO`.

Cluster job files. The `*.sbatch` and `env.sh` files are kept as they were run, with machine-specific values replaced by placeholders: `${REPO}` (the copy of this code used on the cluster), `${SCRATCH}`, `${STORE}`, `${VENV}` (scratch, storage and virtual-environment locations), `${ACCOUNT}`, `${USER}`, `${EMAIL}`, `${HOME}`, `${FLINT_PREFIX}`. In the layout used there the library sat under `${REPO}/code` and each job wrote per-sector records into a `results` directory; those records are the files in `data`. The largest scans took of order 10^2–10^3 core-hours (SU(3) to $j = 30$ about 400, SU(4) to $j = 28$ about 900 and at $j = 30$ about 1500); the one-loop and higher-order SU(3) tests needed nodes with up to 0.5 TB of memory or an H200 GPU.

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